POST MAN

1. What is Postman?

Postman is a popular API testing tool used by developers and testers to send API requests and analyze the responses without writing complex code.

In simple words, Postman helps us test how an API works by sending requests like GET, POST, PUT, DELETE and checking whether the server responds correctly.

It provides a user-friendly interface where users can easily create requests, send data to the server, and view responses in formats like JSON, XML, or HTML.

Postman is widely used in API development, testing, debugging, and automation.

2. Why Postman is Used

Postman is used because it makes API testing easy and efficient.

Main reasons:

- To test APIs without writing code
- To send HTTP requests to the server
- To verify the response from the server
- To debug APIs during development
- To validate API functionality before integration
- To automate API testing

In real software projects, testers use Postman to verify whether APIs are returning correct data, status codes, and response time.

3. Features of Postman

Some important features of Postman are:

1. API Request Creation
Users can create requests like GET, POST, PUT, PATCH, DELETE.

2. Response Validation
Postman shows response body, status code, response time, and headers.

3. Environment Variables
Allows testers to manage different environments such as **development, testing, and production**.

4. API Collections
Requests can be grouped into **collections** for better organization.

5. Automation Testing
Postman allows writing **test scripts using JavaScript**.

6. Collaboration
Teams can share collections and work together.

HTTP Methods in Postman

APIs communicate using **HTTP methods**. The most common ones are:

- GET
- POST
- PUT
- DELETE

These methods define **what action should be performed on the server**.

1. GET Method

What is GET?

GET is used to **retrieve data from the server**.

It does **not modify any data**, it only **reads information**.

Example from your screenshot

GET https://api.restful-api.dev/objects

This request retrieves a list of objects from the API.

What happens when GET is sent

1. The request is sent to the server.
2. The server processes the request.

3. The server returns the requested data.

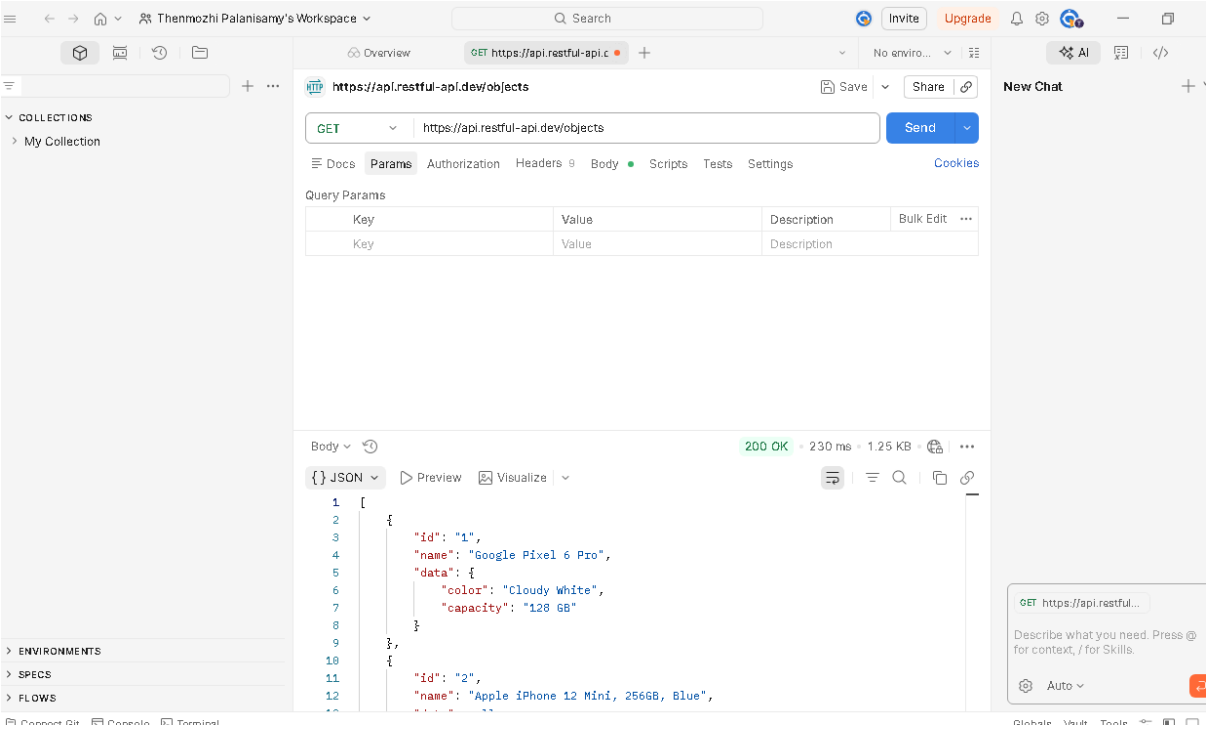
4. The response is displayed in Postman.

Expected Response

• Status Code: **200 OK**
• Response format: **JSON**
• Response time displayed

Example Response:

{
"id": "1",
"name": "Google Pixel 6 Pro",
"data": {
"color": "Cloudy White",
"capacity": "128 GB"
}
}



2. POST Method

What is POST?

POST is used to **create new data in the server**.

When a POST request is sent, **new information is added to the database**.

Example

POST https://api.restful-api.dev/objects

Body (JSON):

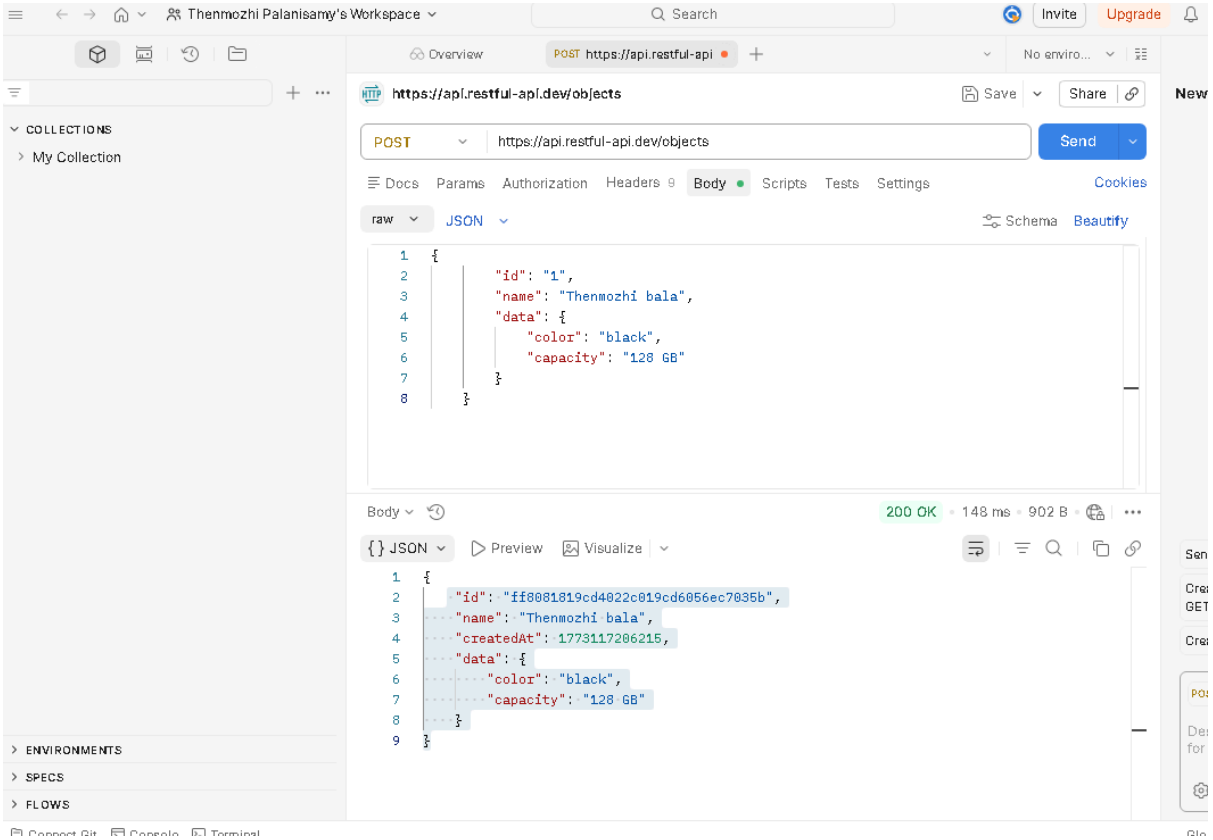
```
{
"name": "Phone",
"data": {
"color": "Black",
"capacity": "128 GB"
}
}
```

What happens when POST is sent

1. The request is sent with a **JSON body**.
2. The server receives the data.
3. The server creates a new record.
4. The server returns confirmation.

Expected Response

• Status Code: **200 OK** or **201 Created**
• New **ID generated**
• Response contains created object



3. PUT Method

What is PUT?

PUT is used to **update an existing resource completely**.

It replaces the **entire object with new data**.

Example from your screenshot

PUT https://api.restful-api.dev/objects

Body:

{
"id": "1",
"name": "Thenmozhi bala",
"data": {
"color": "black",
"capacity": "128 GB"
}

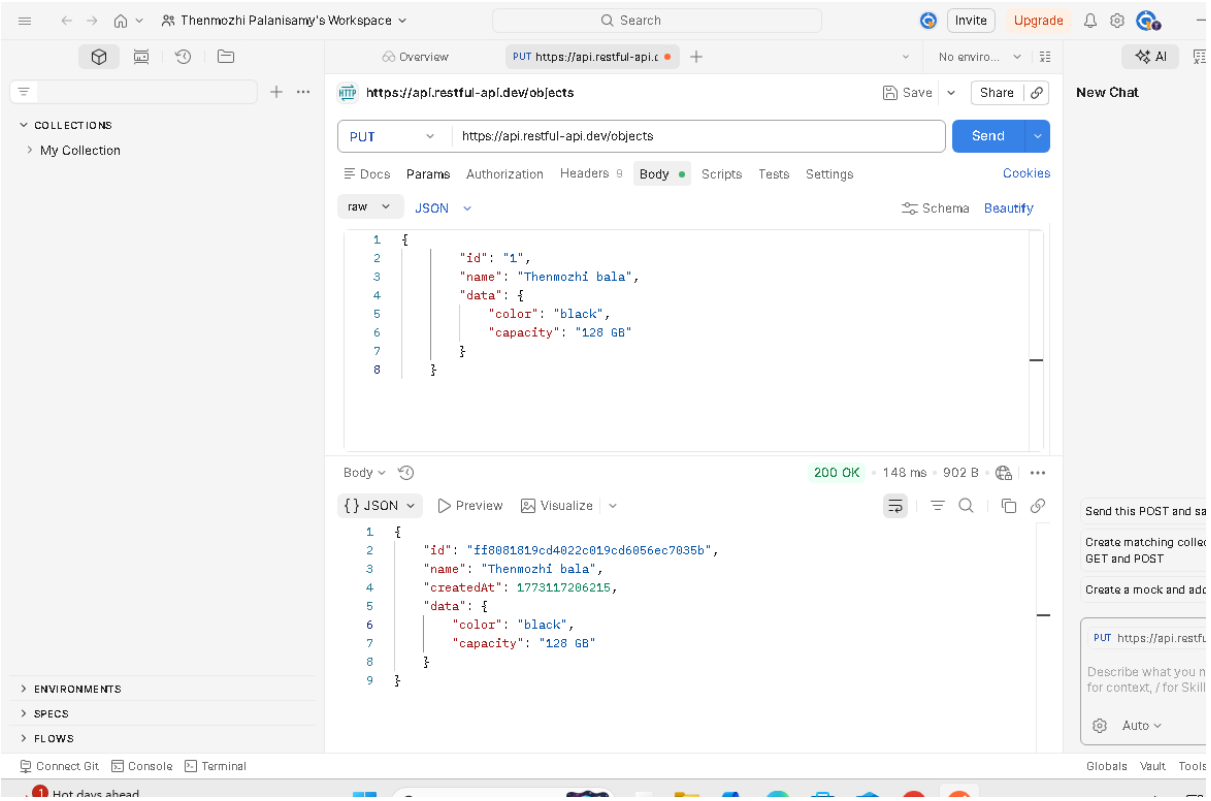
}

What happens when PUT is sent

1. Request is sent with updated data.
2. Server finds the object with that ID.
3. Existing data is replaced with new data.
4. Updated object is returned.

Expected Response

• Status Code: **200 OK**
• Updated object returned in JSON.



4. DELETE Method

What is DELETE?

DELETE is used to **remove data from the server**.

It permanently deletes the specified resource.

Example from your screenshot

DELETE https://api.restful-api.dev/objects/{id}

What happens when DELETE is sent

1. Request is sent with object ID.
2. Server finds that object.
3. Server deletes the record.
4. Server returns confirmation message.

Expected Response

Example:

{
"message": "Object with id = ff8081819cd4022c019cd6056ec7035b has been deleted."
}

Status Code: **200 OK**

Thenmozhi Palanisamy's Workspace

Search

Invite

Upgrad

Overview

DEL https://api.restful-api.d

No enviro...

COLLECTIONS

My Collection

https://api.restful-api.dev/objects/ff8081819cd4022c019cd6056ec7035b

Save

Share

DELETE

https://api.restful-api.dev/objects/ff8081819cd4022c019cd6056ec7035b

Send

Docs Params Authorization Headers 9 Body Scripts Tests Settings

Cookies

raw JSON

Schema Beautify

```
1 {
2     "id": "1",
3     "name": "Thenmozhi bala",
4     "data": {
5         "color": "black",
6         "capacity": "128 GB"
7     }
8 }
```

Body

200 OK 165 ms 866 B

{} JSON Preview Visualize

```
1 {
2     "message": "Object with id = ff8081819cd4022c019cd6056ec7035b has been deleted."
3 }
```

ENVIRONMENTS

SPECS